

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (2 Question × 100 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
Student Name:	S. ROOBINI	Reg. No.:	192421114
Section / Batch:	B. TECH IT	Date:	15-07-26

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
GRAND TOTAL:			100 Marks

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

☉ Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations
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Topic 2: Operational Systems vs DSS

☉ Focus Areas:

- OLTP vs OLAP concepts
- Key differences (data, users, purpose)
- Performance comparison
- Use-case scenarios
- Integration between systems

Topic 3: Architecture of Data Warehouse

🕒 Focus Areas:

- Data warehouse definition
- Three-tier architecture
- ETL process
- Data marts & metadata
- Benefits in decision-making

Topic 4: OLAP and Data Cubes

🕒 Focus Areas:

- OLAP concept
- Data cube structure
- OLAP operations (slice, dice, roll-up, drill-down)
- Applications in analytics

Topic 5: Dimensional Data Modeling

🕒 Focus Areas:

- Fact and dimension tables
- Star schema vs snowflake schema
- Schema design process
- Real-world example

Topic 6: Query Processing & BI Tools

🕒 Focus Areas:

- Query processing stages
- Query optimization
- BI tools overview
- Reporting and visualization
- Examples of open-source BI tools

Topic 7: OLAP Application using KNIME

🕒 Focus Areas:

- Introduction to KNIME
- Workflow components (nodes, connections)
- Performing OLAP operations
- Sample application/demo
- Report generation

Code of ConductIS. ROBINI / 19240114

(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: IS. Robini**RUBRICS FOR CALCULATION**

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors: concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge

Course Coordinator

Head of Department

Signature: _____

Signature: _____

Signature: _____

Date: _____

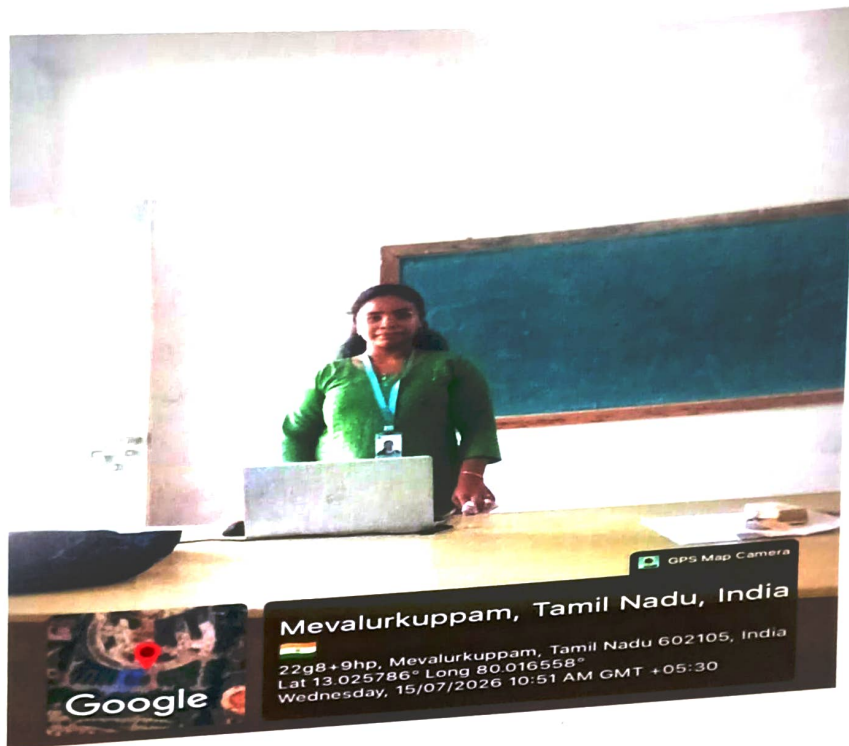
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CSA1617 – DATAWAREHOUSING AND DATAMINING
C01 – ASSESSMENT TOOL – 2 – TECHNICAL PRESENTATION

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ROOBINI

TOPIC 4: OLAP and Data Cubes



OLAP Concept and Data Cube Structure

OLAP (Online Analytical Processing)

Online Analytical Processing (OLAP) is a technology used in data warehousing that enables users to analyze large volumes of data quickly from multiple perspectives. It is designed to support complex analytical queries, helping organizations make informed business decisions. Unlike Online Transaction Processing (OLTP), which manages daily transactions, OLAP focuses on historical and summarized data for analysis and reporting.

Data Cube Structure

A **Data Cube** is the core structure used in OLAP for organizing data in multiple dimensions. It stores information in a way that allows users to analyze data from different viewpoints simultaneously. Each cube consists of **dimensions** and **measures**.

- **Dimensions** are descriptive categories used to organize data, such as **Time, Product, Region, Customer, or Salesperson**.

- **Measures** are numerical values that are analyzed, such as **Sales Amount, Profit, Revenue, or Quantity Sold.**

For example, a retail company may create a data cube using three dimensions:

- **Time** (January, February, March)
- **Product** (Laptop, Mobile, Tablet)
- **Location** (Chennai, Bangalore, Hyderabad)

The measure stored in each cell of the cube is **Sales Amount**. This structure enables users to compare sales across different products, locations, and time periods with ease.

OLAP Operations and Applications in Analytics

1. Slice

The slice operation selects a single value from one dimension, reducing the data cube by one dimension. For example, selecting **Sales for January** displays sales data for all products and locations only during January. It helps users focus on a specific period or category.

2. Dice

The dice operation selects multiple values from multiple dimensions to create a smaller sub-cube. For example, analyzing **Laptop and Mobile sales in Chennai and Bangalore during January and February** provides a more focused view of the data for specific business requirements.

3. Roll-Up

Roll-up summarizes data by moving from detailed information to higher levels of aggregation. For instance, daily sales can be combined into monthly sales, and monthly sales into yearly sales. This operation is useful for generating management reports and identifying overall business performance.

4. Drill-Down

Drill-down is the opposite of roll-up. It allows users to move from summarized data to more detailed information. For example, yearly sales can be expanded into monthly, weekly, and daily sales. This helps managers investigate specific details and identify the causes of business trends.

Applications of OLAP in Analytics

- **Sales Analysis:** Compare sales across products, regions, and time periods to identify top-performing products and market trends.
 - **Financial Analysis:** Monitor revenue, expenses, profits, and budgets to support financial planning and forecasting.
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